Michael C. Burkhart

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Education

Brown University Providence, RI	Ph.D. Applied Mathematics	2013–2019
Rutgers University New Brunswick, NJ	M.Sc. Mathematics	2011–2013
Purdue University West Lafayette, IN	B.Sc.'s Mathematics, Statistics, & Economics	2007–2011

Experience

University of Chicago — Chicago, IL
Sr. Data Scientist — 2024–
- joined the Beaulieu-Jones lab to develop machine learning solutions for healthcare applications

University of Cambridge — Cambridge, UK
Research Associate (Visiting Researcher in 2024) — 2021–2024
- developed multimodal trajectory models for the early diagnosis of neurodegenerative disease
- worked with research engineers at the Alan Turing Institute to automate the detection of covariate shift

Adobe, Inc. — San Jose, CA
Machine Learning Scientist (Sr. in 2021) — 2018–2021
- built and validated predictive models to personalize user experience
- designed and tested personalized pricing interventions within the cancellation flow
- supervised intern projects in representation learning for causal inference and semi-supervised learning

Brown University — Providence, RI
Graduate Research Assistant — 2014–2018
- developed a novel nonlinear filter for online neural decoding
- collaborated with the BrainGate Clinical Trail to test this filter in a brain–computer interface
- experimented with Bayesian solutions to provide robustness against common non-stationarities

Summer research internships at Spotify, U.S.A. (Data Research Intern in New York, NY, 2017) & Argonne National Laboratory (Graduate Research Aide in Lemont, IL, 2012)

Selected Publications

M. Burkhart, B. Ramadan, L. Solo, W. Parker, & B. Beaulieu-Jones, Quantifying surprise in clinical care: detecting highly informative events in electronic health records with foundation models, Pacific Symposium on Biocomputing 31 (2026)

S. Sethi, D. Chen, T. Statchen, M. Burkhart, N. Bhandari, B. Ramadan, & B. Beaulieu-Jones, ProtoECGNet: case-based interpretable deep learning for multi-label ECG classification with contrastive learning, Proceedings of the 10th Machine Learning for Healthcare Conference, PMLR 298 (2025)

M. Burkhart, Actions of nilpotent groups on nilpotent groups, Glasgow Mathematical Journal 67 (2025)

M. Burkhart, Fixed point conditions for non-coprime actions, Proceedings of the Royal Society of Edinburgh Section A: Mathematics 155 (2025)

M. Burkhart, L. Lee, D. Vaghari, A. Toh, E. Chong, C. Chen, P. Tiňo, & Z. Kourtzi, Unsupervised multimodal modeling of cognitive and brain health trajectories for early dementia prediction, Scientific Reports 14 (2024)

M. Burkhart & G. Ruiz, Neuroevolutionary representations for learning heterogeneous treatment effects, Journal of Computational Science 71 (2023)

M. Burkhart, Discriminative Bayesian filtering lends momentum to the stochastic Newton method for minimizing log-convex functions, Optimization Letters 17 (2023)

M. Burkhart & K. Shan, Deep low-density separation for semi-supervised classification, Computational Science – ICCS 2020

M. Burkhart, D. Brandman, B. Franco, L. Hochberg, & M. Harrison, The discriminative Kalman filter for Bayesian filtering with nonlinear and nongaussian observation models, Neural Computation 32 (2020)

D. Brandman, M. Burkhart, J. Kelemen, B. Franco, M. Harrison, & L. Hochberg, Robust closed-loop control of a cursor in a person with tetraplegia using Gaussian process regression, Neural Computation 30 (2018)

D. Brandman, T. Hosman, J. Saab, M. Burkhart, B. Shanahan, J. Ciancibello, …, M. Harrison, J. Simeral, & L. Hochberg, Rapid calibration of an intracortical brain computer interface for people with tetraplegia, Journal of Neural Engineering 15 (2018)

See https://burkh4rt.github.io/pubs for a complete list

Dissertation

M. Burkhart, A Discriminative Approach to Bayesian Filtering with Applications to Human Neural Decoding, Ph.D. advised by Prof. Matthew Harrison, Division of Applied Mathematics, Brown University, Providence, 2019

Patents

M. Burkhart & G. Ruiz, Causal inference via neuroevolutionary selection, U.S. Patent Application #17/748,891 filed 2022, granted 2026 as US 12,602,589 B2

M. Burkhart & K. Shan, User classification from data via deep segmentation for semi-supervised learning, U.S. Patent Application #16/681,239 filed 2019, granted 2022 as US 11,455,518 B2

M. Burkhart & K. Modarresi, Digital experience enhancement using an ensemble deep learning model, U.S. Patent Application #16/375,627 filed 2019, granted 2023 as US 11,816,562 B2

Selected Preprints

I. Lee, L. Solo, M. Burkhart, B. Ramadan, W. Parker, & B. Beaulieu-Jones, Representation before training: a fixed-budget benchmark for generative medical event models, arXiv:2604.16775

L. Solo, M. McDermott, W. Parker, B. Ramadan, M. Burkhart, & B. Beaulieu-Jones, Efficient generative prediction for EHR foundation models: the SCOPE and REACH estimators, arXiv:2602.03730

N. Hahn, E. Stein, BrainGate Consortium (128 named members including M. Burkhart, D. Brandman, E. Brown, & M. Harrison), J. Donoghue, J. Simeral, L. Hochberg, & F. Willett, Long-term performance of intracortical microelectrode arrays in 14 BrainGate clinical trial participants, medRxiv:2025.07.02.25330310

M. Abroshan, M. Burkhart, O. Giles, S. Greenbury, Z. Kourtzi, J. Roberts, M. van der Schaar, J. Steyn, A. Wilson, & M. Yong, Safe AI for health and beyond – monitoring to transform a health service, arχiv:2303.01513

M. Burkhart, Linear transformations & the multivariate generating function, arXiv:1209.3357

Selected Talks

Local conjugacy and primary-type decompositions in nonabelian cohomology, Algebra Seminar, Mathematics Institute, University of Warwick, UK, 2025

Clustering trajectories of neurodegenerative disease, Trustworthy AI for medical & health research workshop, Cavendish Laboratory, Cambridge, UK, 2022

Neuroevolutionary feature representations for causal inference, International Conference on Computational Science (ICCS), London, UK, 2022

Community Involvement

Cambridge Psych. Dept. Cambridge, UK	**Research Staff Representative**	2022–2023
ICCS Conference	**Program Committee Member** • thematic track on Applications of Computational Methods in Artificial Intelligence and Machine Learning	2019–2021
Brown SIAM Student Chapter Providence, RI	**Vice President, Chapter Records** **Interdepartmental Liaison Officer** • organized events within the applied math community	2015–2017
Rutgers Math Dept. New Brunswick, NJ	**Graduate Liaison Committee Member**	2012–2013
Purdue Student Publishing Foundation West Layfayette, IN	**Member, Corporate Board of Directors** **Chairperson, Finance Committee** • oversaw the *Exponent*, Purdue's Independent Daily (at the time) Student Newspaper	2009–2011

Online

Homepage • LinkedIn • Google Scholar • Github • OrcID

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